A PARTIAL RESULT ON THE DAVIS-WIELANDT INDEX

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Source Question

Bhunia, Sain and Paul introduced the Davis-Wielandt index

$$\eta_{dw}(X) = \inf\{dw(T) : T \in S_{\mathcal{L}(X)}\}$$

of a normed linear space X. At the end of arXiv:2006.15323 they ask for separate complete characterizations of the properties

$$\eta_{dw}(X) = 1 \qquad \text{and} \qquad \eta_{dw}(X) = \sqrt{n(X)^2 + 1},$$

where $n(X)$ is the numerical index.

Result

We record a concrete family for the first property.

Theorem 1. *Let $X = (\mathbb{R}^2, \|\cdot\|_p) = \ell_p^2$ be real. If $2 \le p \le 8$, then*

$$\eta_{dw}(X) = 1.$$

Proof Intuition

The Davis-Wielandt index is always at least 1. Thus it is enough to find one norm-one operator whose Davis-Wielandt radius is 1. On ℓ_p^2, the natural candidate is the rank-one nilpotent $T(a, b) = (0, a)$. At the point $(1, 0)$, T attains its norm and the unique norming functional annihilates $T(1, 0)$, exactly as for a skew-symmetric Hilbert-space operator. The only issue is to control all other unit vectors; this reduces to a one-variable inequality with a sharp endpoint at $p = 8$.

Scalar Inequality

Lemma 1. *For $1 \le r \le 4$ and $0 \le s \le 1$,*

$$s(1 - s^r)^{2-1/r} + s^2 \le 1.$$

Proof. If $1 \le r \le 2$, then $s^r \ge s^2$, hence

$$(1 - s^r)^{2-1/r} \le 1 - s^r \le 1 - s^2,$$

because $2 - 1/r \ge 1$. Multiplying by $s \le 1$ gives the result.

It remains to consider $2 \le r \le 4$. Put $a = 1/r$, so $a \in [1/4, 1/2]$, and set $y = s^r$. The desired inequality is

$$H_a(y) := 1 - y^{2a} - y^a(1 - y)^{2-a} \ge 0.$$

We first note that $H_a(y)$ is increasing in a on $[1/4, 1/2]$. Indeed,

$$\frac{\partial H_a(y)}{\partial a} = 2y^{2a} \log(1/y) - y^a(1 - y)^{2-a} \log \frac{y}{1 - y}.$$

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For $0 < y \le 1/2$ this is non-negative. For $1/2 < y < 1$, write $\rho = (1-y)/y \in (0,1)$. Since $2a \le 1$ and $2 - a \ge 3/2$,

$$\frac{\partial H_a(y)}{\partial a} \ge \frac{2\log(1+\rho) - \rho^{3/2}\log(1/\rho)}{(1+\rho)^2}.$$

Now $2\log(1+\rho) \ge \rho$, while $\rho^{3/2}\log(1/\rho) \le \rho$, since $\sqrt{\rho}\log(1/\rho) \le 2/e < 1$. Hence the derivative is non-negative.

It is therefore enough to prove the endpoint $a = 1/4$:

$$y^{1/4}(1-y)^{7/4} + y^{1/2} \le 1.$$

Writing $v = y^{1/4}$, this is equivalent to

$$v(1-v^4)^{7/4} \le 1 - v^2.$$

After raising to the fourth power and putting $u = v^2$, it suffices to show

$$P(u) := u^2(1-u)^3(1+u)^7 \le 1, \qquad 0 \le u \le 1.$$

The logarithmic derivative of P is

$$\frac{2}{u} - \frac{3}{1-u} + \frac{7}{1+u} = \frac{2 + 4u - 12u^2}{u(1-u)(1+u)}.$$

Thus the only interior maximum occurs at $u = (1+\sqrt{7})/6$, and direct substitution gives $P(u) < 0.619 < 1$. Since $P(0) = P(1) = 0$, the endpoint case and the lemma follow. $\square$

Proof of the Theorem

Let $T : \ell_p^2 \to \ell_p^2$ be

$$T(a,b) = (0,a).$$

Clearly $\|T\| = 1$. Since $p > 1$, each non-zero point of ℓ_p^2 has a unique norming functional. If $x = (a,b) \in S_{\ell_p^2}$, then

$$x^{\#} = (\operatorname{sgn}(a)|a|^{p-1}, \operatorname{sgn}(b)|b|^{p-1}) \in S_{(\ell_p^2)^*}$$

is the unique element of $\mathcal{J}(x)$. Therefore

$$|x^{\#}(Tx)|^2 + \|Tx\|^4 = |a|^2|b|^{2p-2} + |a|^4.$$

Put $s = |a|^2$ and $r = p/2$. Since $2 \le p \le 8$, we have $1 \le r \le 4$, and since $|a|^p + |b|^p = 1$,

$$|a|^2|b|^{2p-2} + |a|^4 = s(1-s^r)^{2-1/r} + s^2 \le 1$$

by the lemma. Hence $dw(T) \le 1$. At $x = (1,0)$, the unique norming functional is $x^{\#} = (1,0)$, and

$$|x^{\#}(Tx)|^2 + \|Tx\|^4 = 0 + 1 = 1,$$

so $dw(T) = 1$.

The source paper proves $\eta_{dw}(X) \ge 1$ for every normed space X. Since $T \in S_{\mathcal{L}(X)}$ and $dw(T) = 1$, we conclude

$$\eta_{dw}(\ell_p^2) = 1.$$